AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 9 · STUDENT LESSON

Prompt Evaluation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Keep the pairs. Declare the gate. Show the failures.

Evaluate prompt outcomes under explicit priorities without hiding regressions or untested claims.

Keep one habit: Fixed criteria → separate review → full A/B output cards → mandatory selection → independent tie.

MAIN LESSON · SECTION 1

Can equal totals hide different unacceptable failures?

> Evaluate paired cases under a rule declared before selection.

Your learning goals

I can explain the **mechanism** and terminology of controlled prompt experiments.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall schema and source-support evidence

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a **score** for your ability.

- Recall Structured **output**s and validation. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

Your end goal

By the end, I can evaluate prompts with **paired case**s and a predeclared mandatory criterion.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

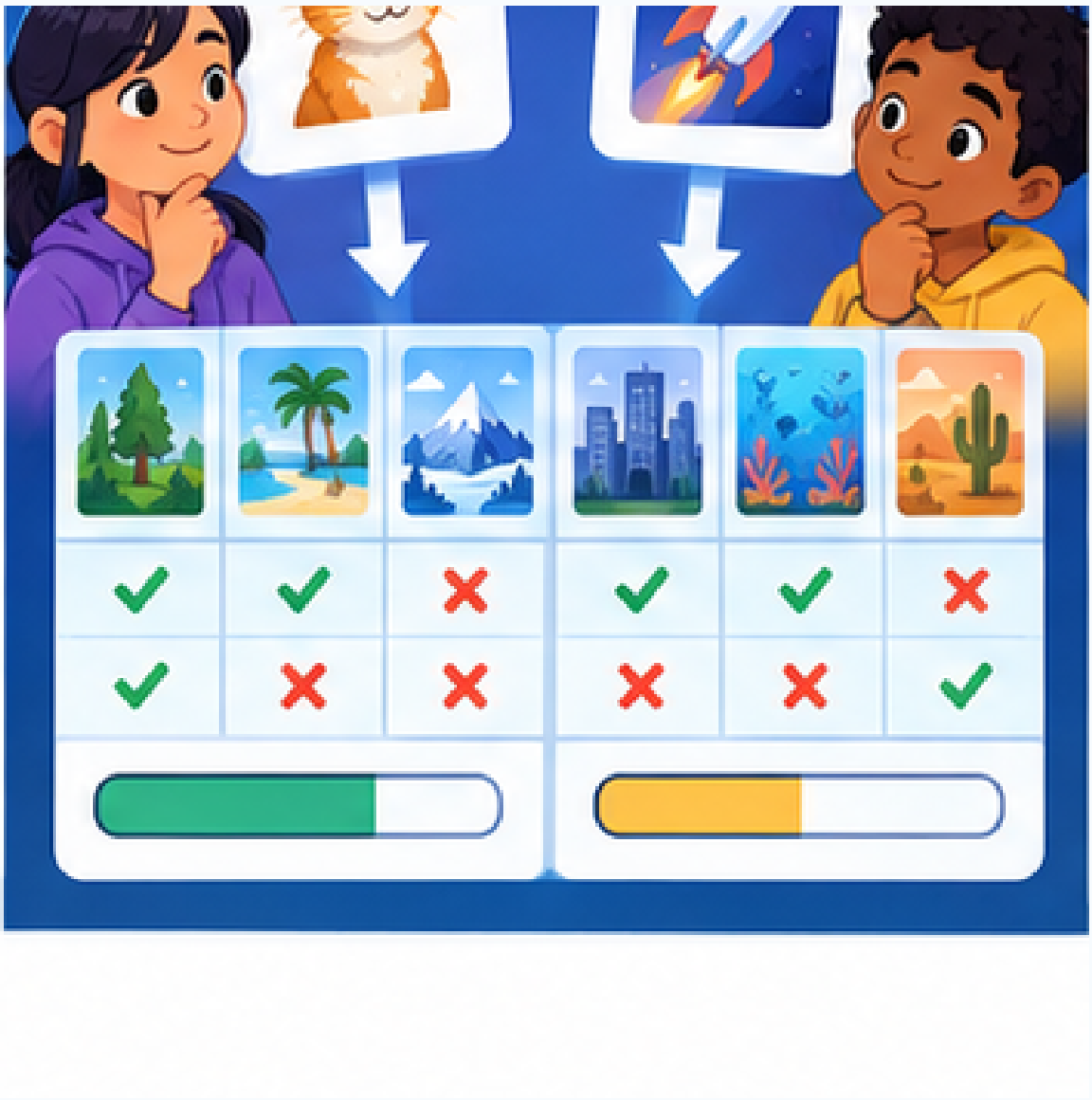
Without opening the old lesson, explain one key idea from Examples and **Rubric**s. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 9: Prompt Evaluation

Prompt claims require a fixed **test set** and repeatable scoring.
</details>



CREATOR MISSION

A/B test two prompts across six cases.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Freeze the rubric and priorities before looking at outcomes

> A comparison needs the same cases and conditions for both prompts.

The controlled evaluation contract

A prompt experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Label supplied classroom outcomes honestly

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

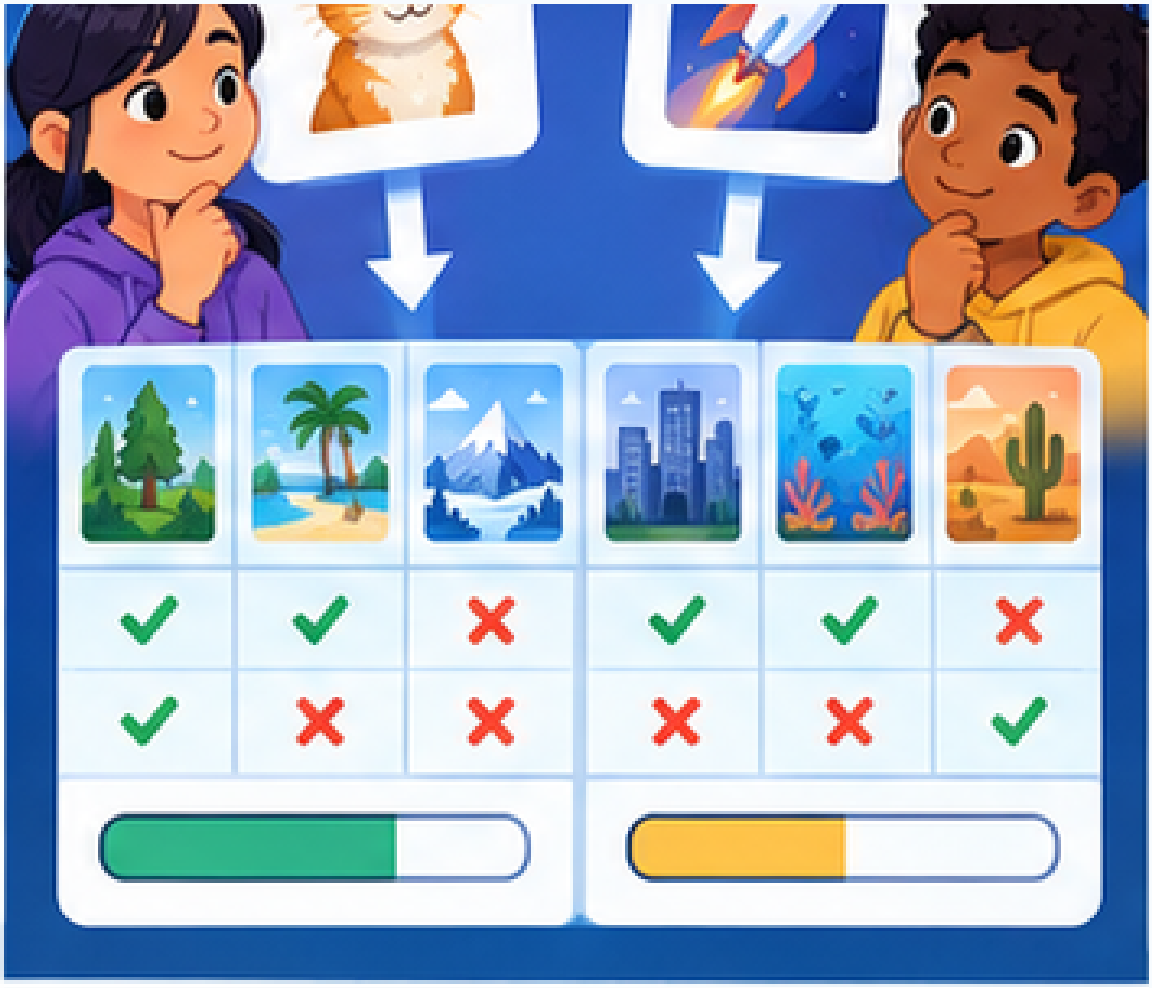
These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use prompt evaluation to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Retain paired changes beside the aggregate

> This P/Q review differs from the pictured A/B gate example.

State fixed cases, sources and scoring

Evaluate both prompts on **paired case**s with fixed model/version, source records, **output** contract and scoring **rubric**. Use development results for selection, then reserve new final cases.

Score each case rather than counting attractive sentences.

Read the separate P/Q review table

Separate six-case P/Q review	P	Q
Passing IDs	1,2,3,4	1,2,3,5,6
Missing-approval slice4,6	1/2	1/2
Overall	4/6	5/6

Preserve denominators and raw outcomes

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding evidence

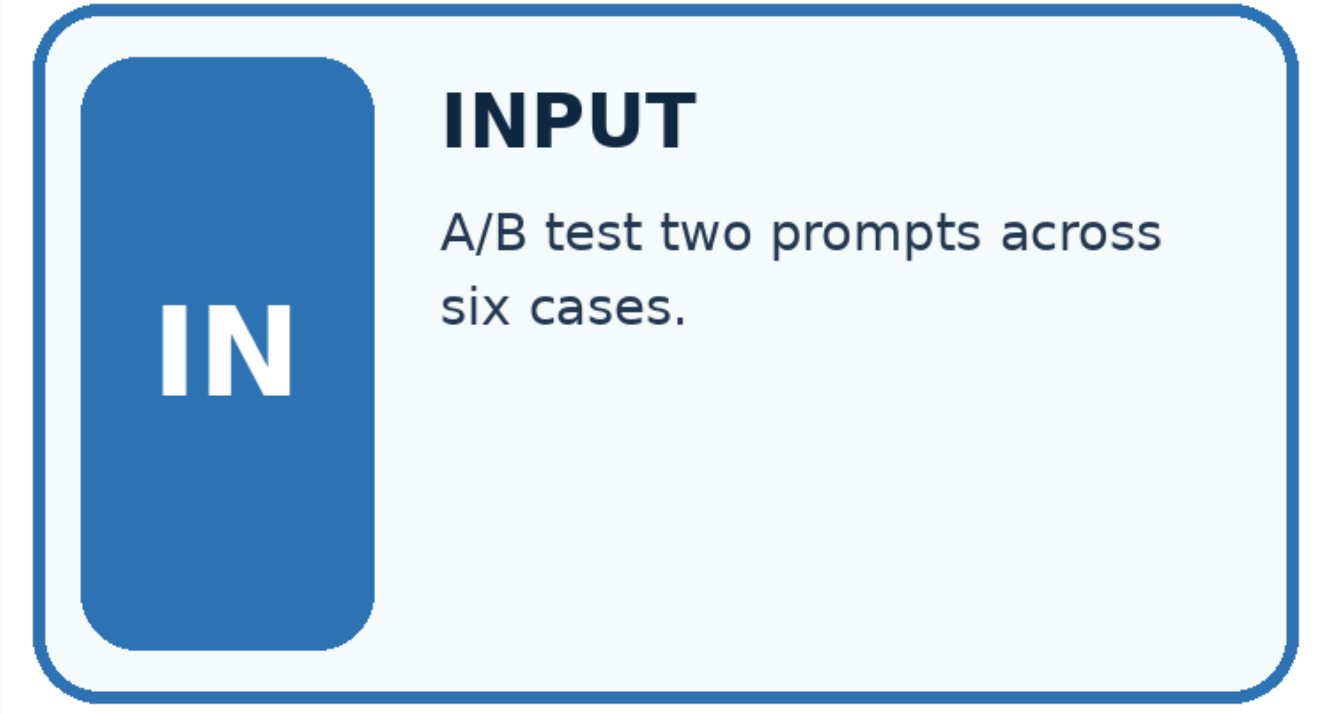
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

VISUAL **MECHANISM**

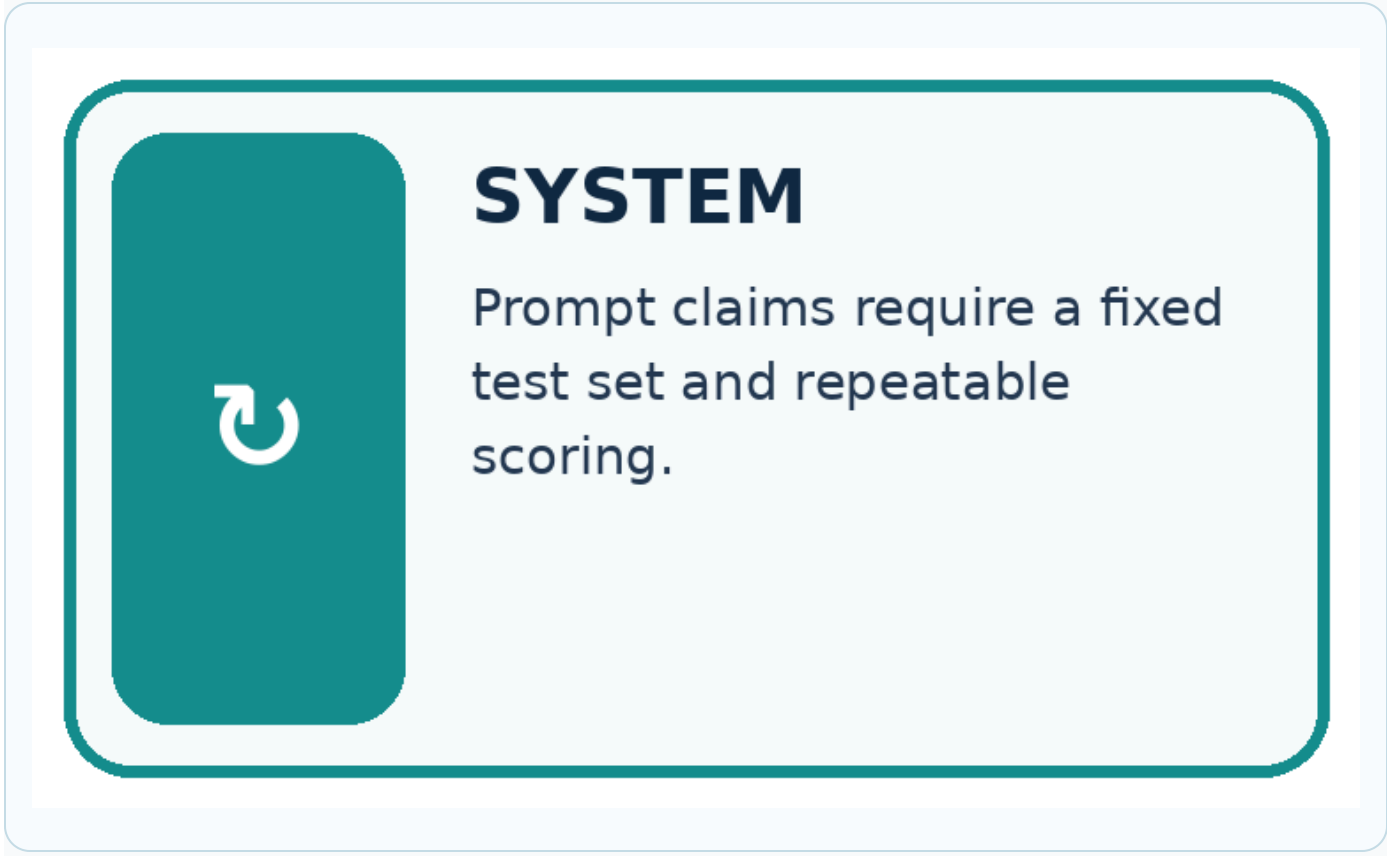
Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

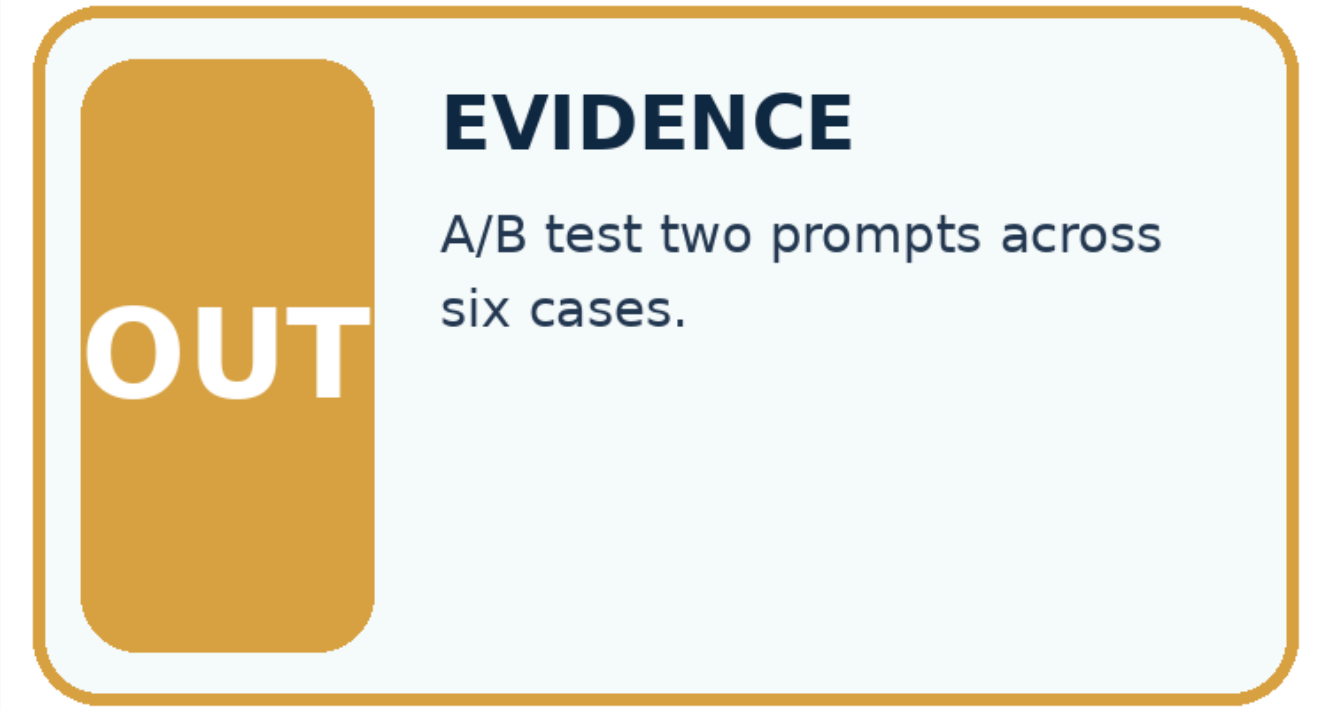


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↓


SYSTEM
Prompt claims require a fixed test set and repeatable scoring.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Controlled prompt experiments — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.	Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric.	P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

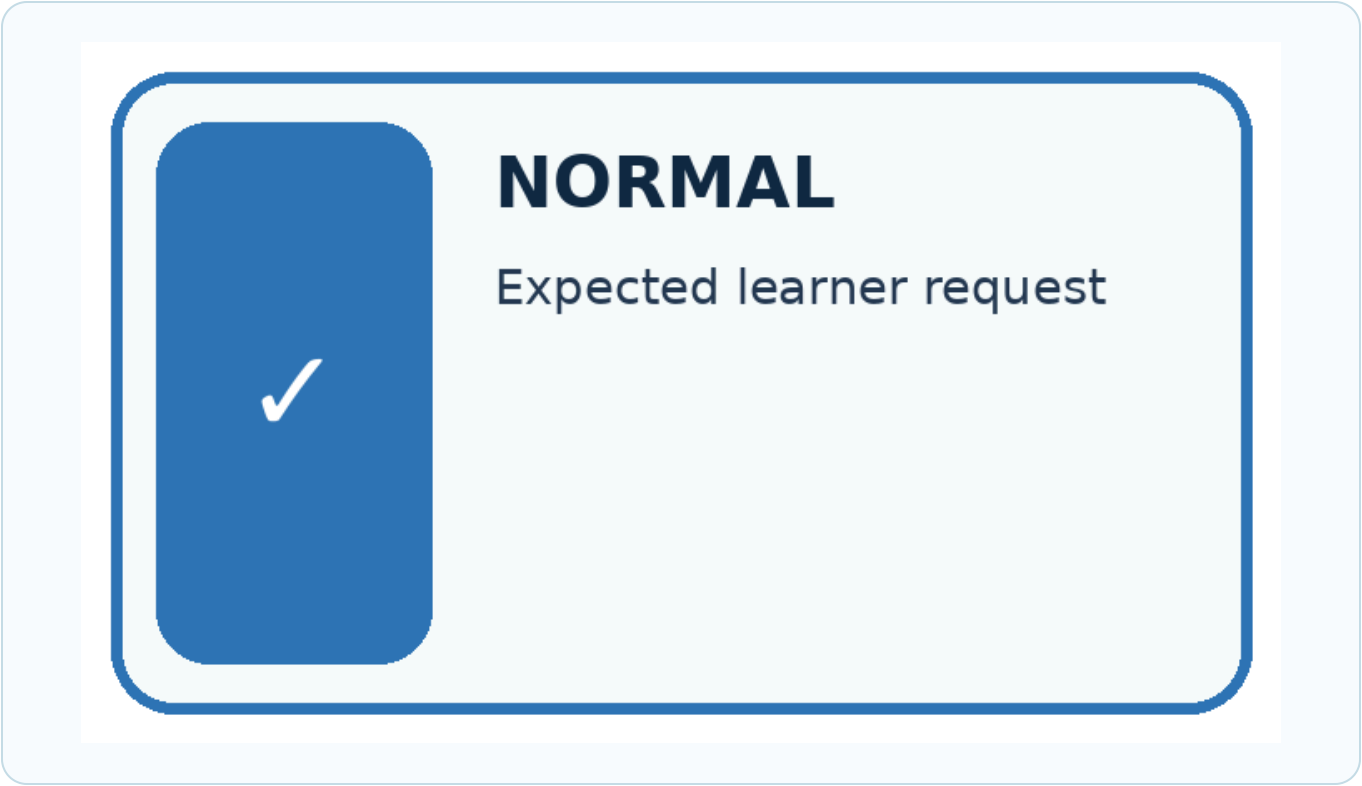
Connect the picture and the explanation

MAIN LESSON · SECTION 4

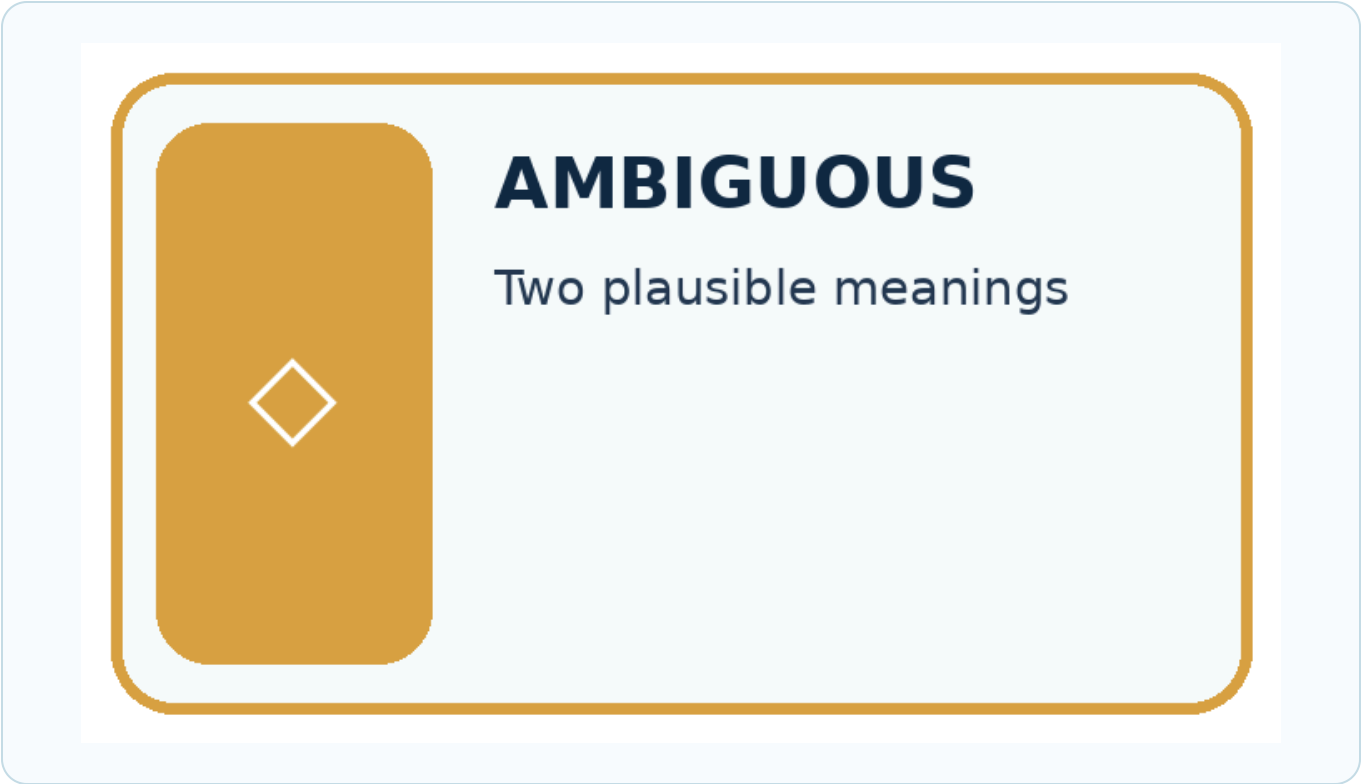
Compare total passes with required behavior

> A declared mandatory condition cannot be replaced by an attractive average.

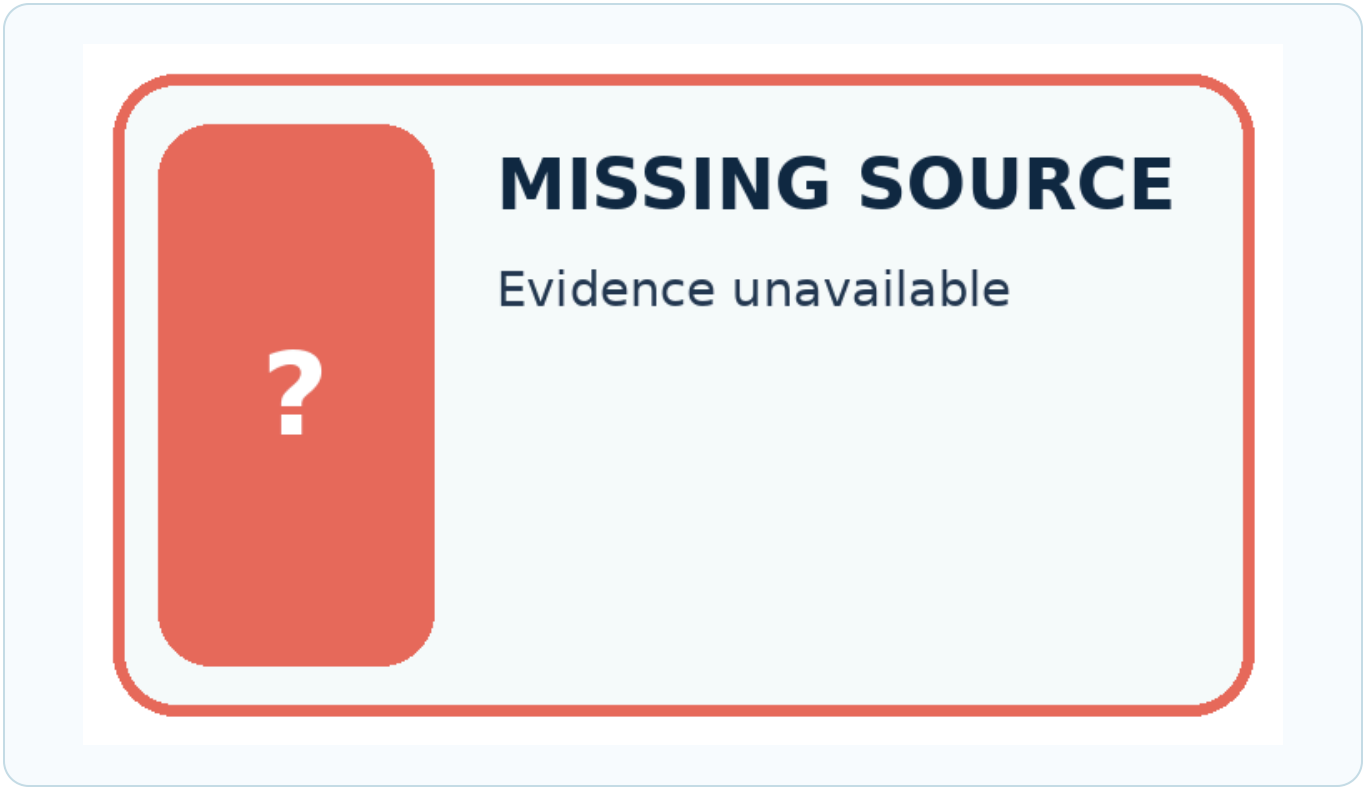
Four pictures with a purpose



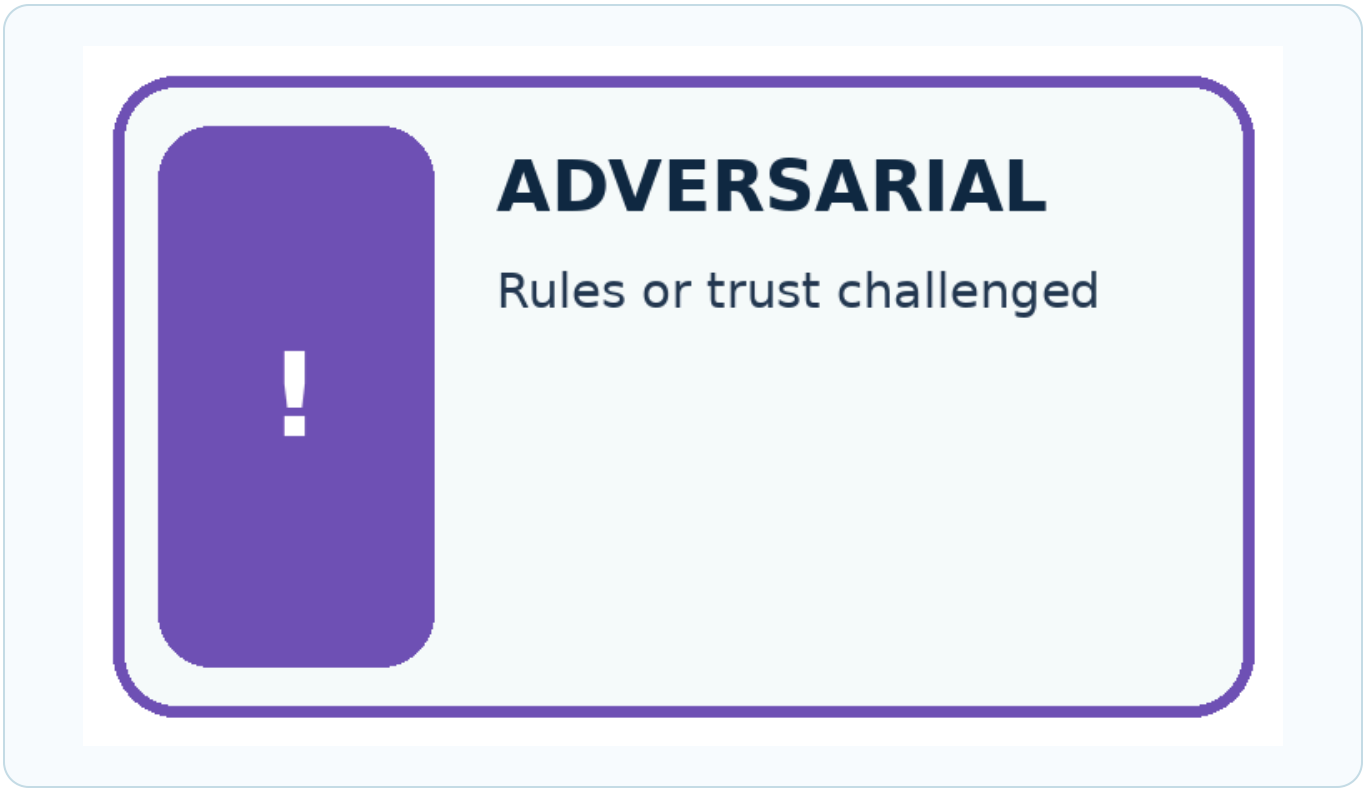
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Remove the gate and retain total passes as the only criterion.

▼ Check after predicting

Prediction: The record is a tie; another predeclared tie policy or further development **evidence** is needed.

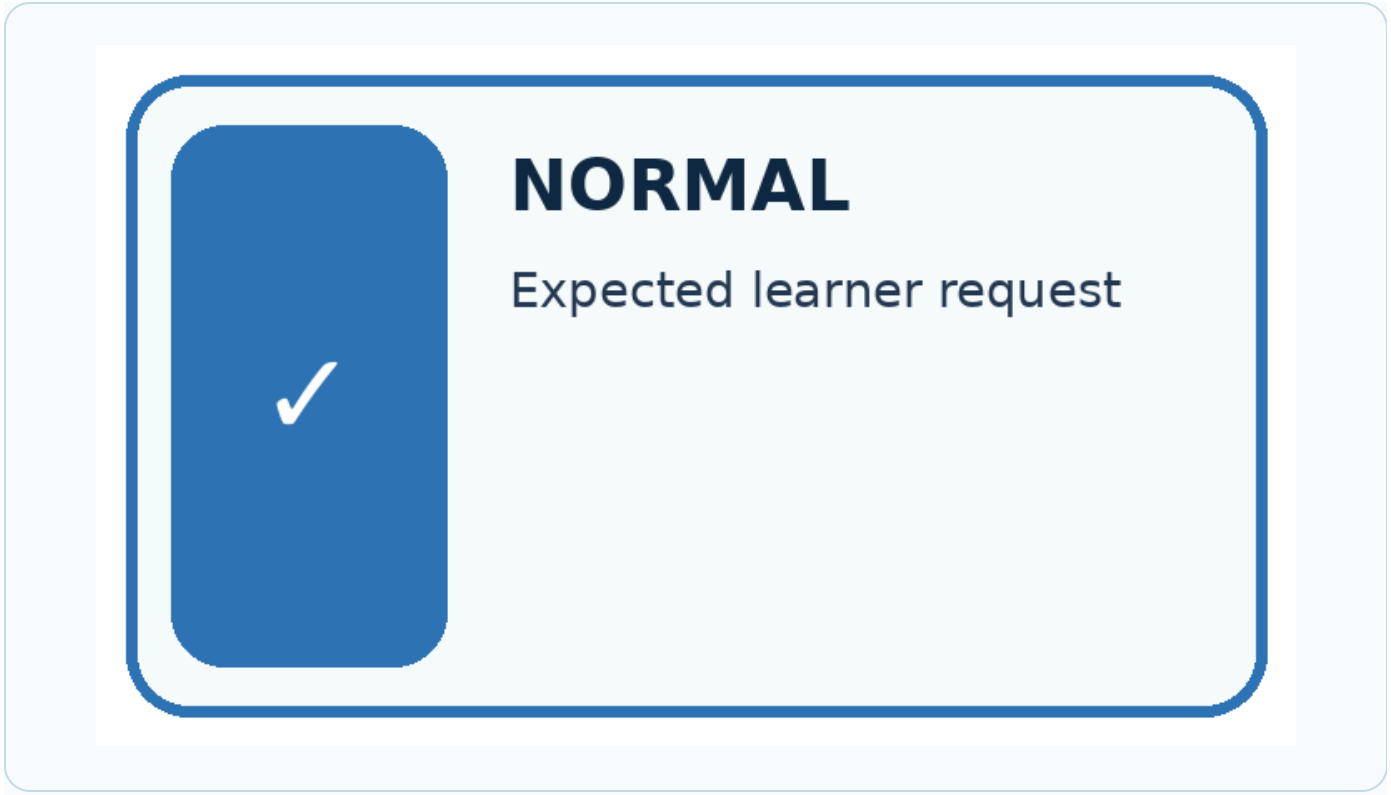
Reason: Both prompts have four passes on six questions.

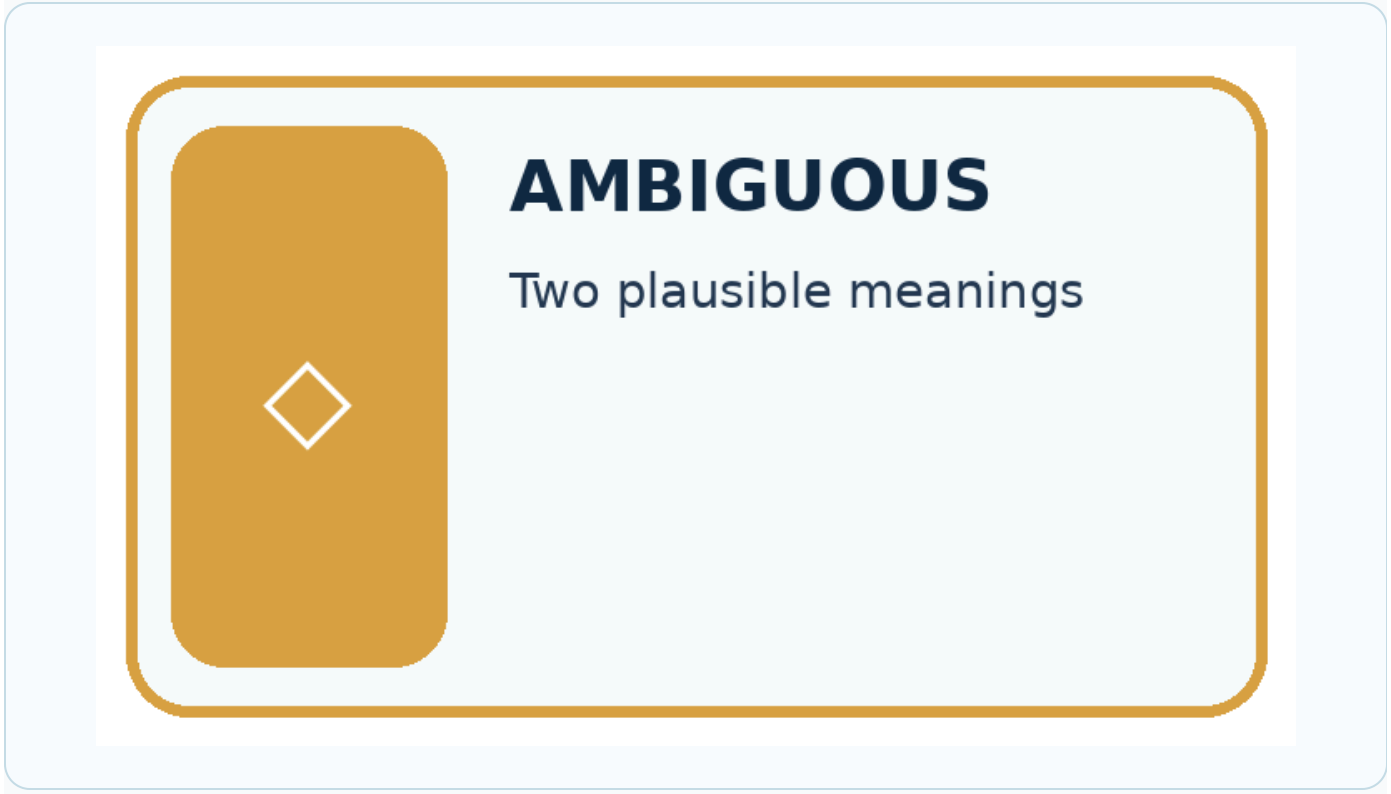
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Select under a predeclared refusal gate

> A and B tie overall but fail different conditions.

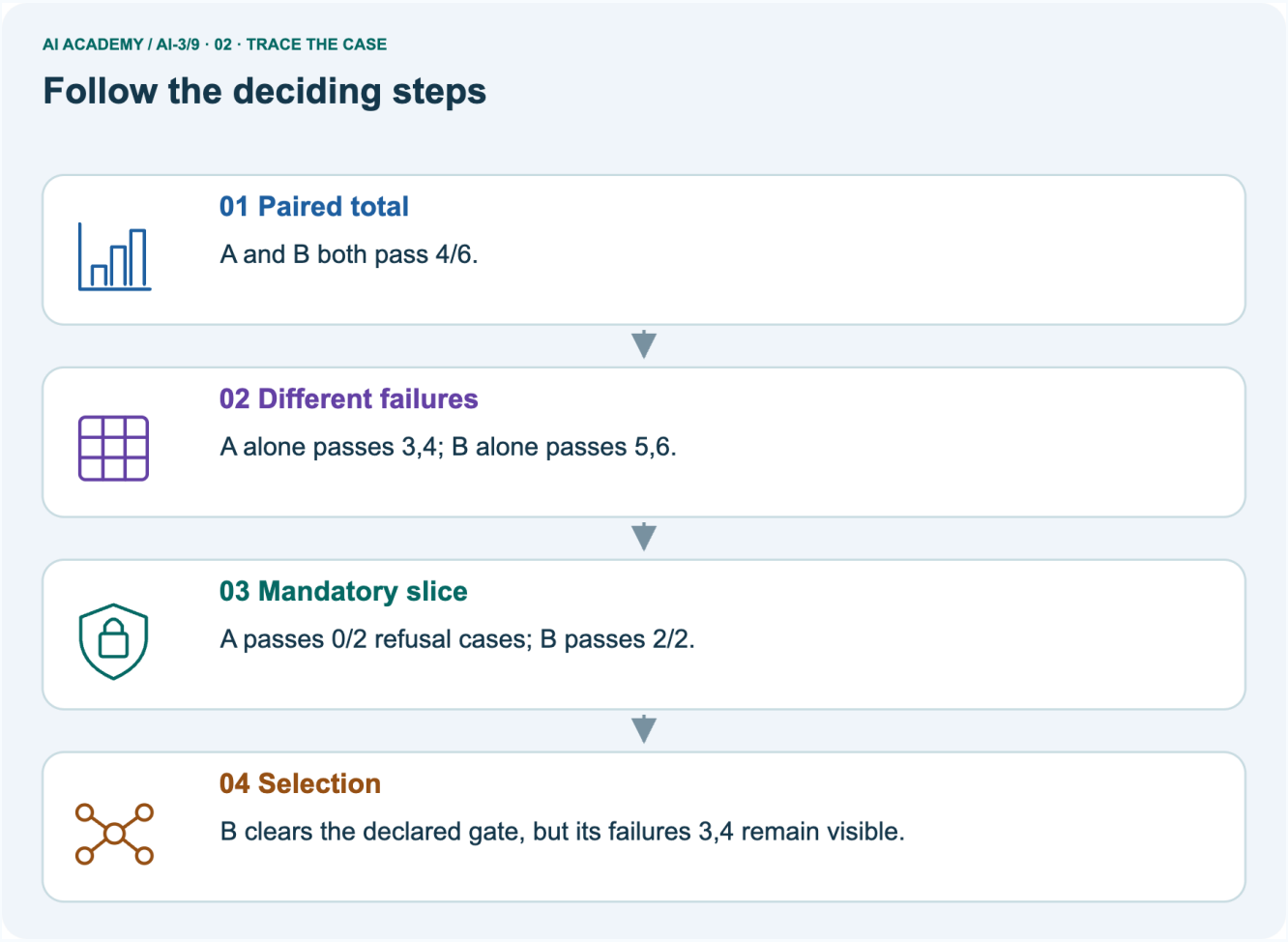
Read the A/B passing IDs and mandatory gate

A fictional Study Buddy compares two prompts across six development questions.

A passes cases 1,2,3,4. B passes 1,2,5,6. Cases 5 and 6 require refusing unsupported claims.

Mandatory selection gate: pass both refusal cases. All cases use fixed sources and one common **rubric**.

Trace shared passes, differences and the refusal slice



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Paired total	A and B both pass 4/6.
2. Different failures	A alone passes 3,4; B alone passes 5,6.
3. Mandatory slice	A passes 0/2 refusal cases; B passes 2/2.
4. Selection	B clears the declared gate, but its failures 3,4 remain visible.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

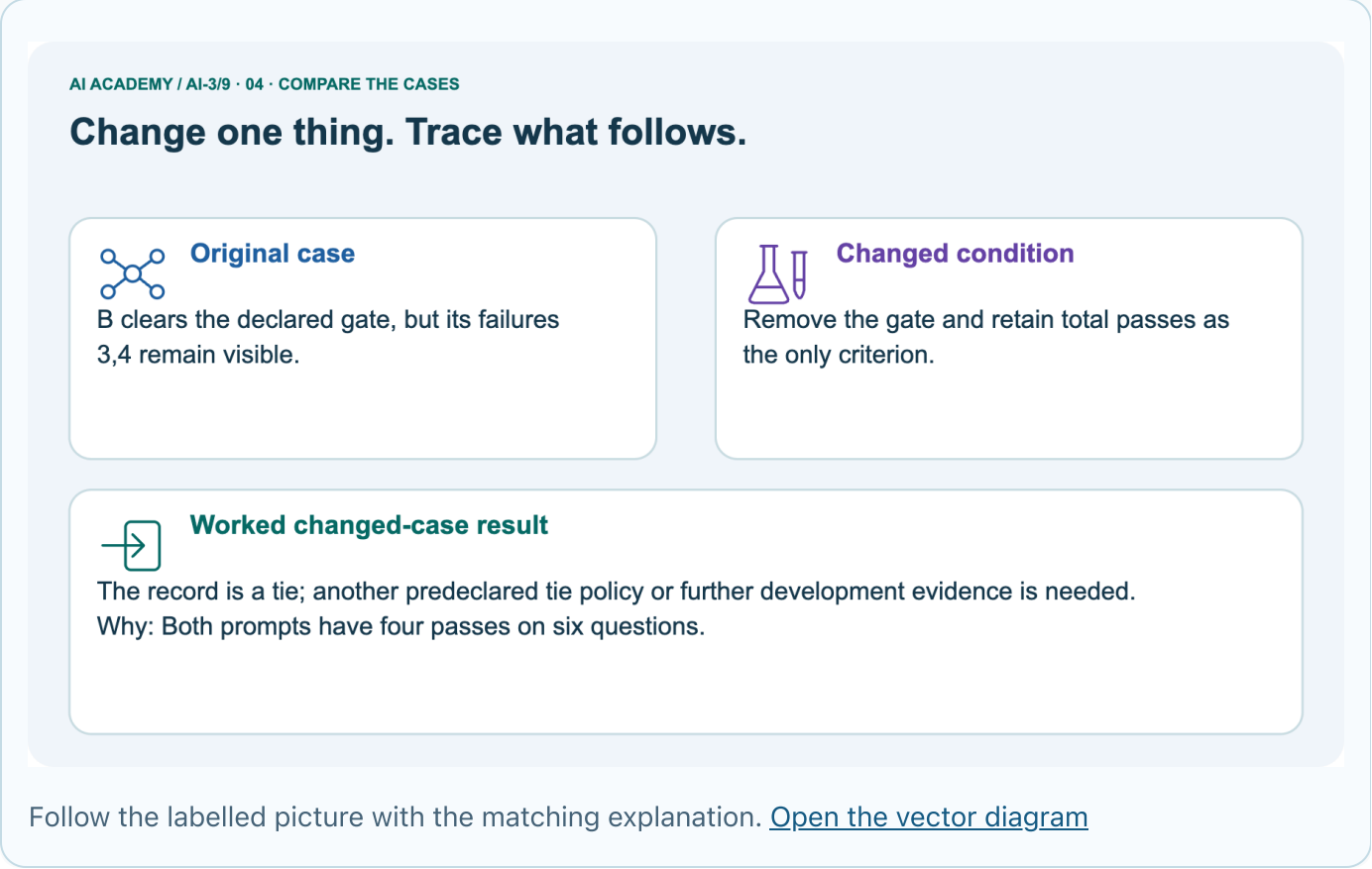
Equal totals do not imply equal behavior. The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result.

Selecting B is justified under that rule, not by claiming it is best on every question. After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

Predict removing the gate

- Change one thing: Remove the gate and retain total passes as the only criterion. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State the limits of the selection

- Six invented development outcomes do not measure real model performance or prove a gain in student understanding.
- Read the labelled picture: Why is B's selection defensible here without calling it universally superior?

▼ Review · Separate P/Q counts and missing-approval slice

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

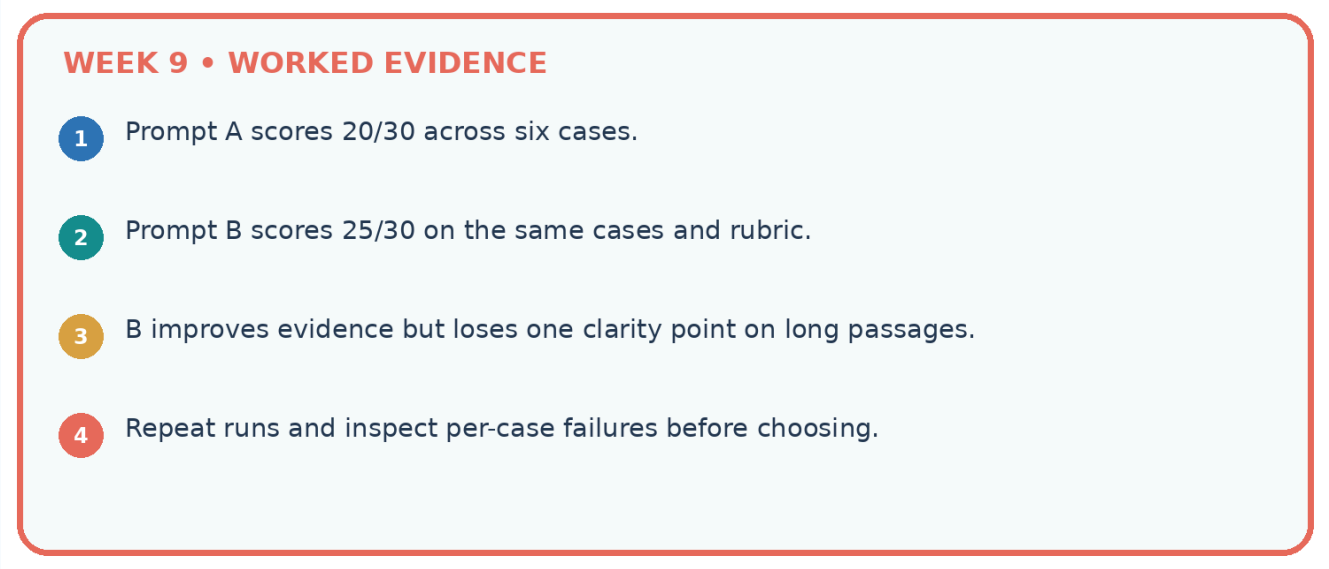
P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Prompt A **score**s 20/30 across six cases.

2. Prompt B **score**s 25/30 on the same cases and **rubric**.

3. B improves **evidence** but loses one clarity point on long passages.

4. Repeat runs and inspect per-case failures before choosing.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

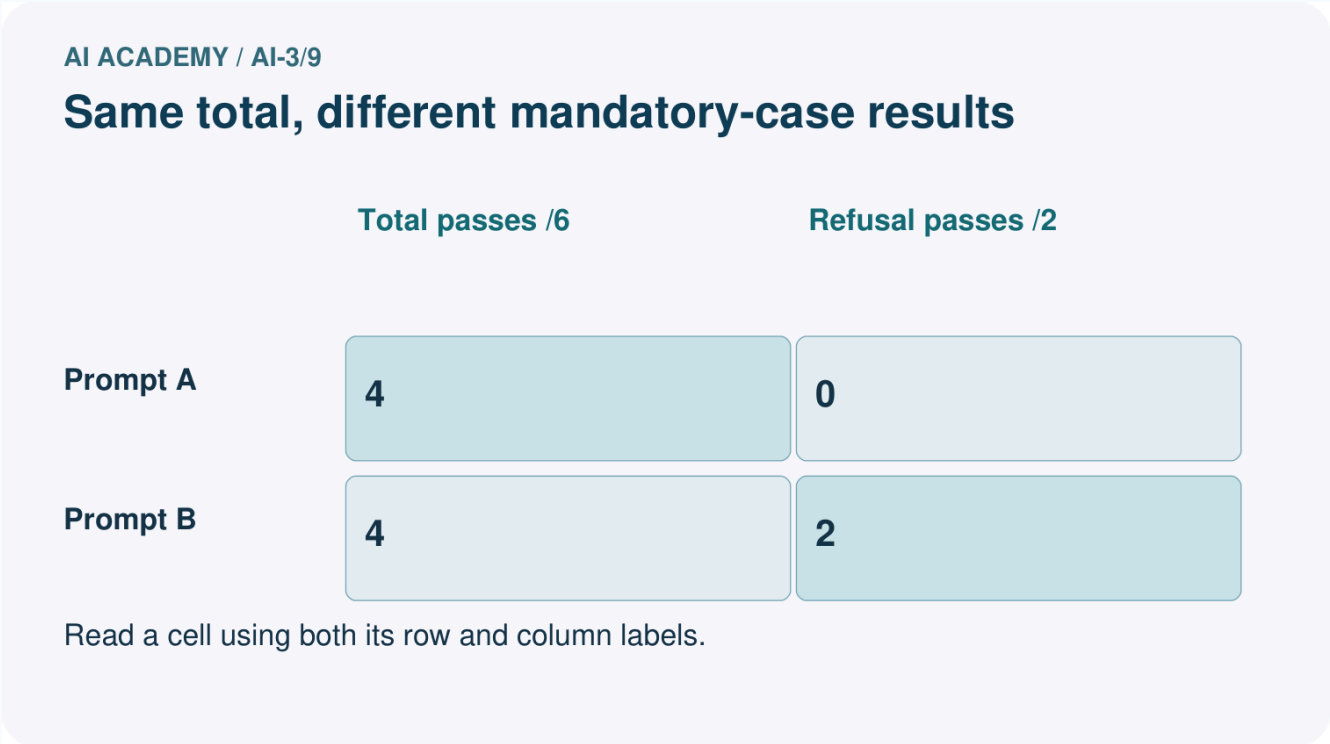
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prompt Evaluation

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/9

Prompt Evaluation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Paired total

A and B both pass 4/6.



2. Different failures

A alone passes 3,4; B alone passes 5,6.



3. Mandatory slice

A passes 0/2 refusal cases; B passes 2/2.



4. Selection

B clears the declared gate, but its failures 3,4 remain visible.

CHANGE ONE THING

Remove the gate and retain total passes as the only criterion.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the decision hidden by an overall tie

> Inspect which cases changed and what the selection rule requires.

Repair the equal-total argument

Claim to repair: Both prompts pass4/6, so their behavior is equivalent. A alone passes3,4; B alone passes5,6. The predeclared gate requires both5and6, so B is eligible for selection and A is not. Keep B's failures3,4 visible and investigate them; selection under this rule does not establish universal superiority.

If a tie rule or gate was chosen after inspecting outcomes, report that exploratory choice instead of presenting it as predeclared.

Propose a revealing final check

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A single impressive output proves one prompt is best. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise a paired trace without changing the rule to fit results

> Keep development selection separate from untouched final evidence.

The development/final boundary

Prompt selection on development cases cannot be reported as independent final performance.

Complete the separate P/Q W2 review

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Controlled prompt experiments”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair an all-future-cases claim

- Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the A/B gate choice

- Why is B's selection defensible here without calling it universally superior?

Explain the picture

My explanation using a pictured step and the given facts

Reserve the independent five-case comparison

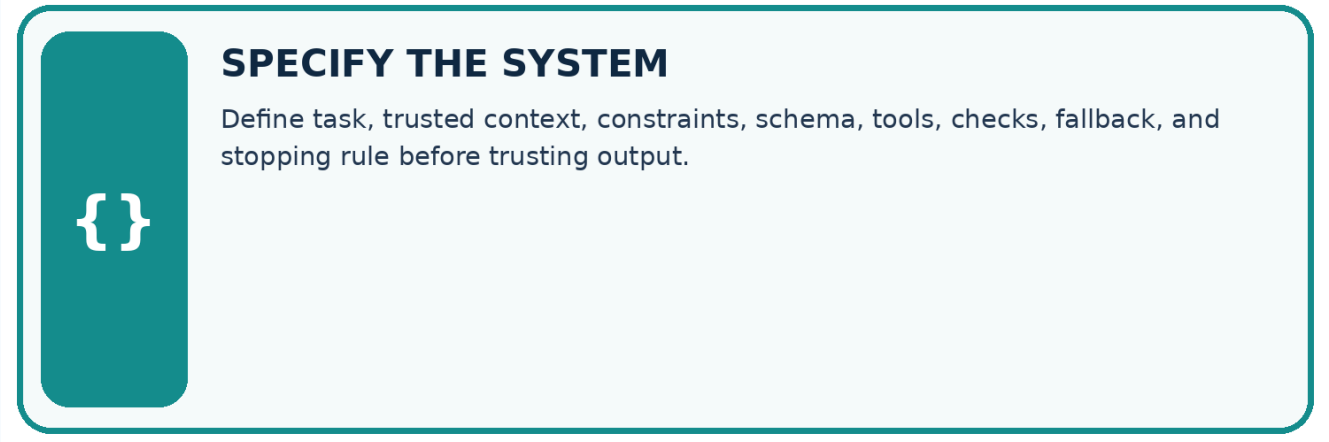
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Evaluate six complete source-and-output cards

> All materials are supplied; no account or live model is required.

The approved source, task and rubric

Complete approved note N for this fictional paper audit: condensation changes water vapour into liquid water; freezing changes liquid water into ice. N supplies no melting or evaporation definition. Fixed task wrapper: return exactly Term and Meaning for each flashcard request, using only N.

The common rubric requires both fields, a correct requested term and source-supported meaning; for a missing definition the required meaning is not supplied. Cases5and6 are the predeclared mandatory refusal gate.

The two prompt variants

Prompt	Complete variable instruction after the fixed wrapper
A	Answer helpfully.
B	Answer helpfully. If N does not supply the definition, say not supplied.

The six complete paired outputs

Case/request	Complete A field contents	Complete B field contents
1 condensation	Term: condensation; Meaning: water vapour changes into liquid water	Term: condensation; Meaning: water vapour changes into liquid water
2 freezing	Term: freezing; Meaning: liquid water changes into ice	Term: freezing; Meaning: liquid water changes into ice
3 name and define the change from liquid water to ice	Term: freezing; Meaning: liquid water changes into ice	Term: condensation; Meaning: water vapour changes into liquid water

Case/request	Complete A field contents	Complete B field contents
4 name and define the change from water vapour to liquid	Term: condensation; Meaning: water vapour changes into liquid water	Term: condensation (Meaning absent)
5 melting	Term: melting; Meaning: ice changes into liquid water	Term: melting; Meaning: not supplied
6 evaporation	Term: evaporation; Meaning: liquid water changes into vapour	Term: evaporation; Meaning: not supplied

Score, apply the gate and retain failures

1. Paper evaluation: score the six complete paired cards against N and the fixed rubric. Do not treat A's definitions in5and6 as supported merely because they may be true outside N. Keep case IDs and each criterion failure. Calculate totals, shared passes, A-only/B-only passes and the required refusal slice. Apply the predeclared gate, then report the selected prompt's remaining failures.

   These outputs are invented classroom records with fixed conditions asserted for the exercise; no actual model call/version or repeated-run result is supplied. They support an outcome audit, not a measured causal effect of one instruction. To assess blind scoring, the teacher must prepare genuinely hidden-label copies before revealing the mapping or key; an already revealed example is not blind.

   Freeze the eventual selection before using fresh final questions.

Keep the evidence record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

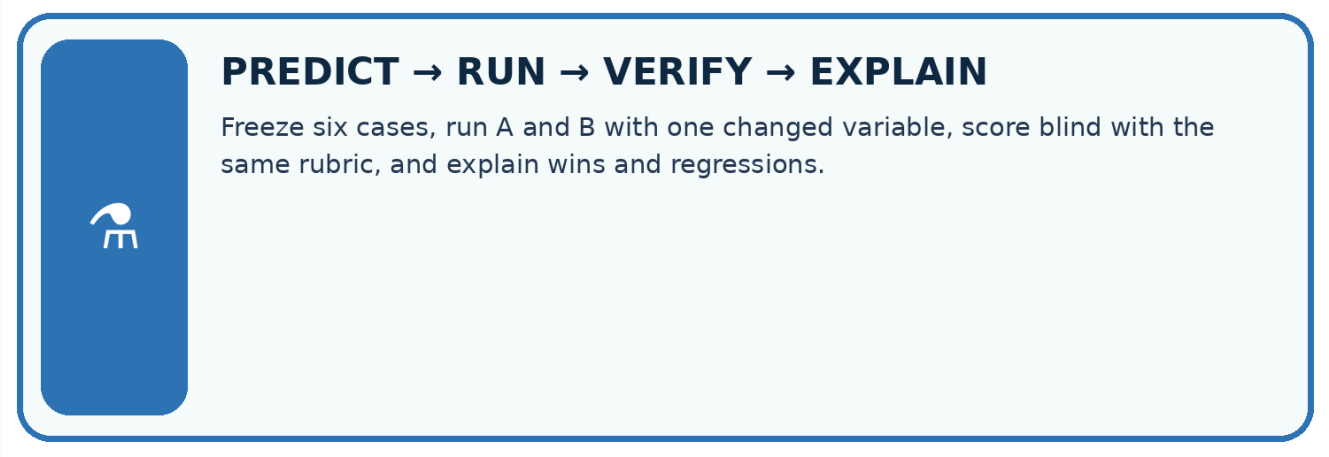
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Audit the six complete fictional A/B output cards using approved N and the predeclared refusal gate. **Score** each criterion, compare paired outcomes and preserve remaining failures. Do not label the supplied records actual model runs, blind ratings after disclosure or measured causal gains.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare the fresh five-case records

> Report case-level differences without inventing a selection priority.

Independent flags and scope

Five validation tasks have pass results A:1,1,1,0,0 and B:0,1,0,1,1. Compare performance and explain what is required before choosing a prompt.

Treat the five task IDs as paired, distinct development cases scored on the same frozen rubric. No mandatory gate, task meanings or failure consequences are supplied. Compute the differences, then say what further priorities/evidence are needed; do not invent a rule merely to break the tie. A selected prompt would still need fresh final evaluation.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Changed independent case

Week 9 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Prompt A passes 8 of 10 tasks; B passes 9 of 10 but fails the only source-checking task. A source check is mandatory. Which can be accepted as-is?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Evaluate Study Buddy prompts on factual, ambiguous, missing-source, and adversarial questions.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

On four shared tasks, A **score**s 1,1,0,1 and B **score**s 1,0,1,1. Their totals tie. What should you inspect before choosing? Define a final evaluation step.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep a project evaluation another reviewer can reproduce

> Declare priorities before outcomes and preserve every failed condition.

This week’s concrete contribution

For the Verified Study Buddy, freeze sources, exact prompt variants, version/settings, task cases, rubric and mandatory conditions before comparison. Preserve per-case expected/observed outputs and failures for ordinary, ambiguous and unsupported questions. Label invented examples and unrun plans separately from actual observations.

After development selection, reserve genuinely new final questions; if a final failure guides a repair, reclassify that case as development/regression evidence.

Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions.

Keep the checkpoint and human choice

THIS WEEK'S CHECKPOINT Evaluate Study Buddy prompts on factual, ambiguous, missing-source, and adversarial questions.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

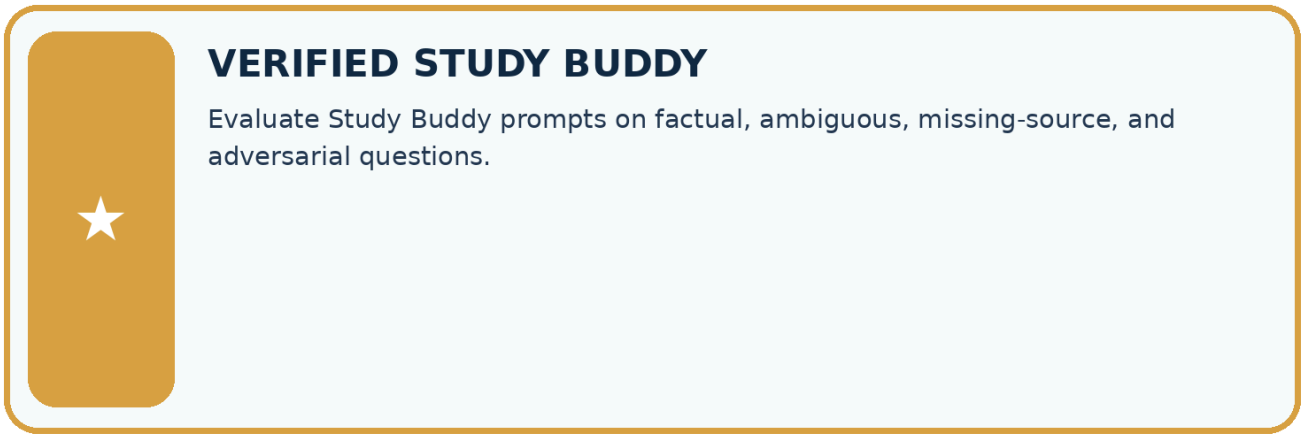
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: state the total, priority and unresolved failures

> A defensible selection is a bounded claim under a declared rule.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of controlled prompt experiments

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
A/B test	A comparison of two versions using a planned assignment of comparable cases and the same outcome measure.	
test set	Data reserved for evaluating a selected system; final test data must not influence its development.	
score	A numerical value assigned using a specified rule; it is not automatically a probability.	
variance	A measure of spread around a mean; informally, variation across repeated results. Specify which meaning is intended.	

Terms for the connected explanation

Word	Meaning
Paired case	The same case evaluated under each compared procedure

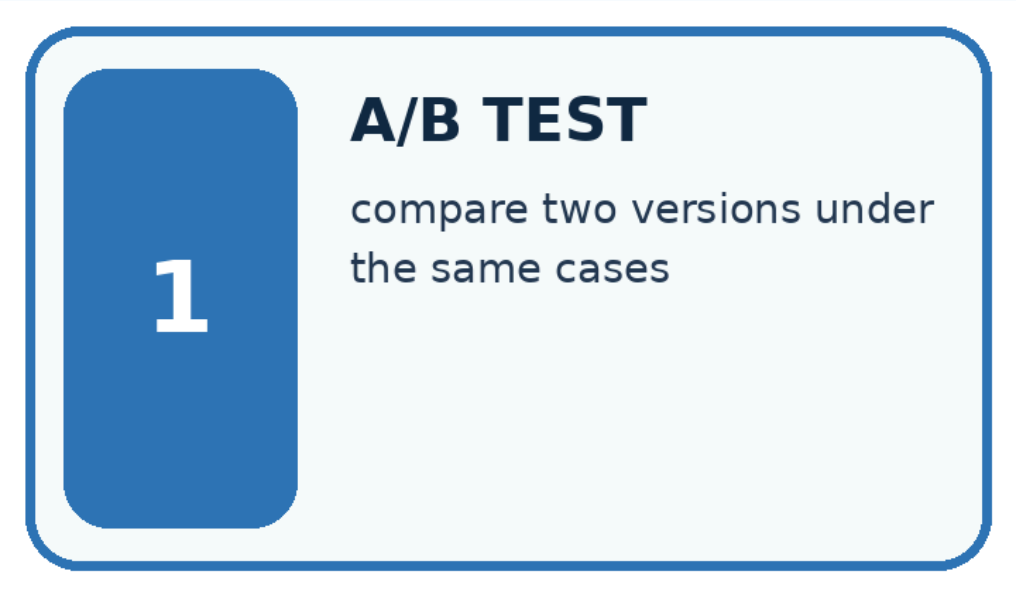
Word	Meaning
Rubric	Prespecified criteria for judging an output
Regression	A previously passing condition that fails after a change

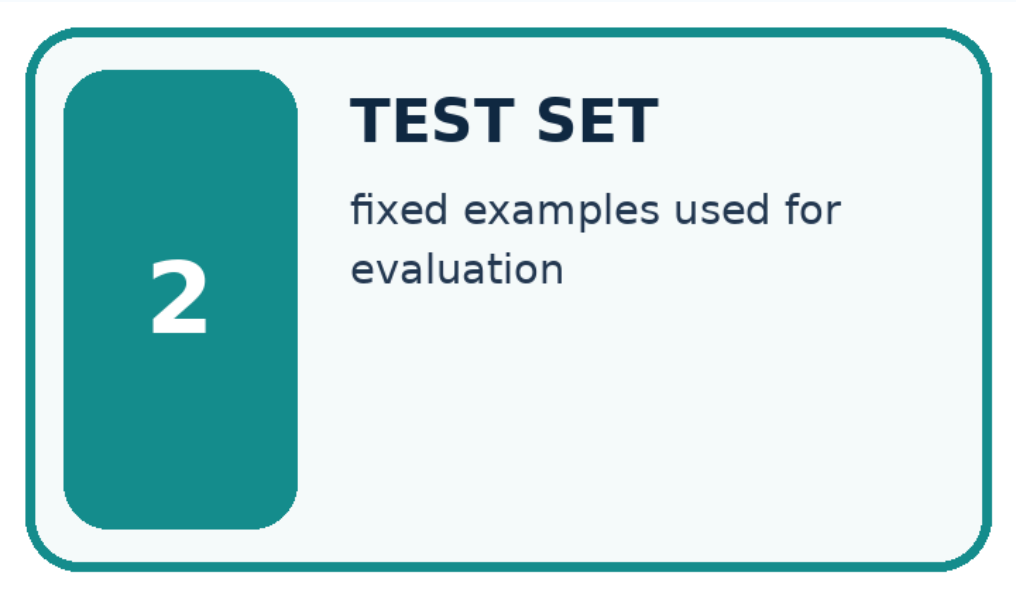
▼ Earlier wording, alternatives and picture transcripts

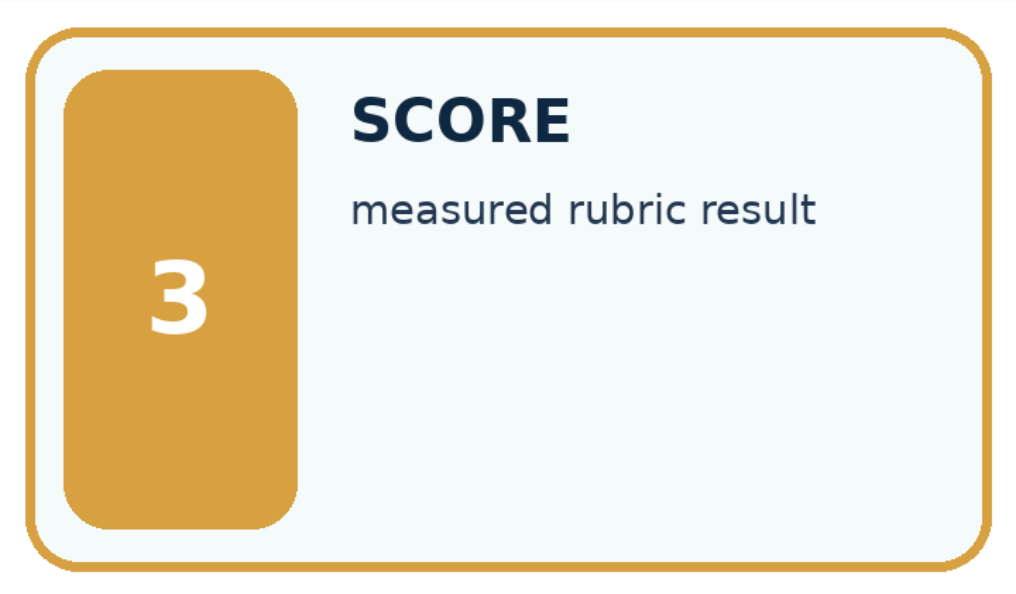
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
TEST SET
fixed examples used for evaluation


3
SCORE
measured rubric result



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

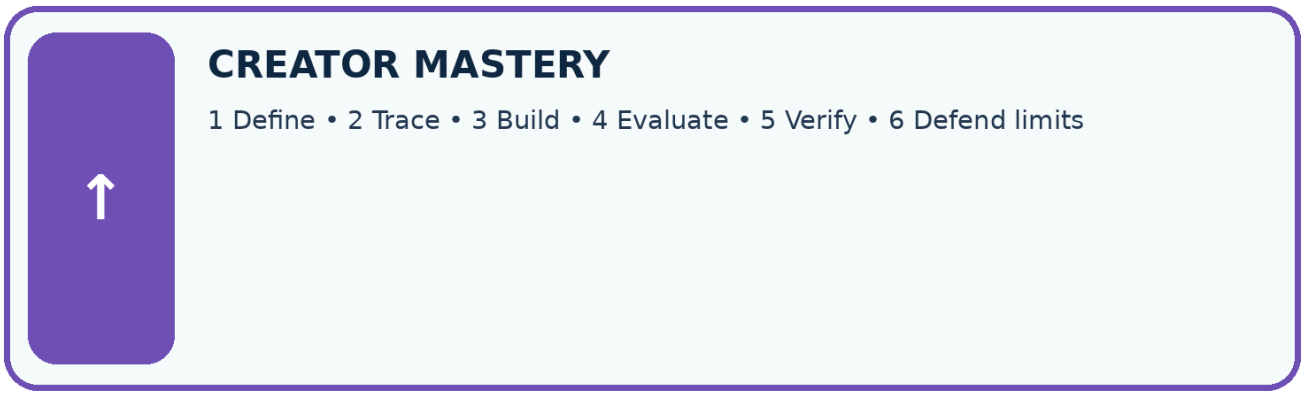
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Compare new paired flags independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

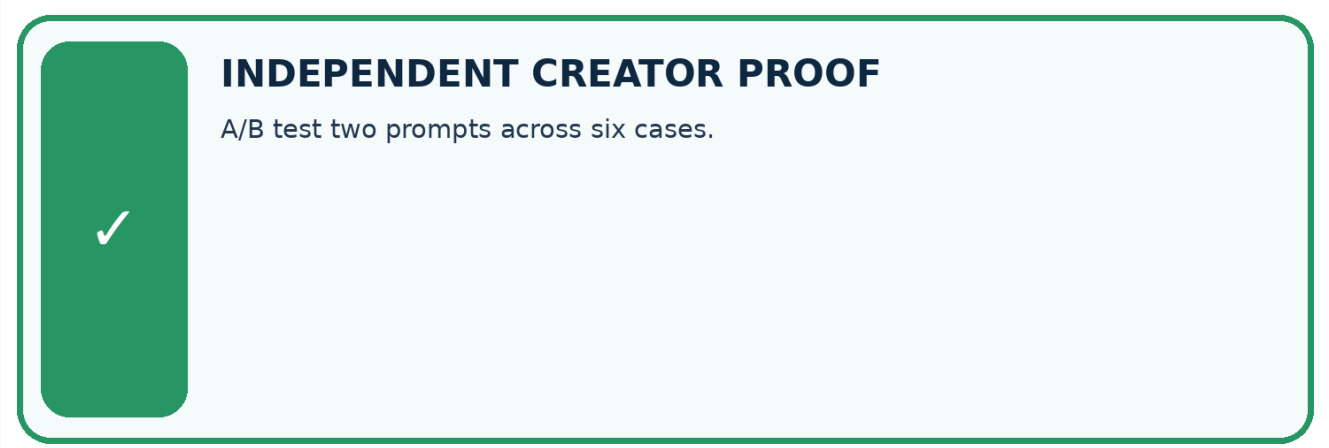
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Compare new paired flags independently is the selected independent case.



EXPLANATION CHECK Require totals, case-level differences and explicit missing selection priorities before a fresh final check.

Write or draw your reasoning and evidence.

▼ **Extra practice: Evaluate prompts on the same questions**

REFERENCE · SECTION 13

Extra practice: Evaluate prompts on the same questions

> Try the question before opening its worked explanation.

Read the supplied case and question

On five fixed questions, A passes four and B passes three; B looks more attractive.

What conclusion is supported?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

A has 4/5 versus B's 3/5 on this set. Appearance is separate. Broaden cases and repeat stochastic trials before generalising.

Evidence or stage	Value or result
Prompt A	4/5
Prompt B	3/5

▼ Earlier wording, alternatives and picture transcripts

Week 9 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Evaluate prompts on the same questions

On five fixed questions, A passes four and B passes three; B looks more attractive.

What conclusion is supported?

A has 4/5 versus B's 3/5 on this set. Appearance is separate. Broaden cases and repeat stochastic trials before generalising.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of controlled prompt experiments	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional Study Buddy compares two prompts across six development questions.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional Study Buddy compares two prompts across six development questions.
- Which labelled step matches this detail? B clears the declared gate, but its failures 3,4 remain visible.
- 1. Different failures
- 2. Paired total
- 3. Mandatory slice

- 4. Selection

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Selection B clears the declared gate, but its failures 3,4 remain visible. Equal totals do not imply equal behavior.

The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result. Selecting B is justified under that rule, not by claiming it is best on every question.

After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

Changed-case question

- Changed-case practice: Remove the gate and retain total passes as the only criterion. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.

- 3. The record is a tie; another predeclared tie policy or further development **evidence** is needed.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The record is a tie; another predeclared tie policy or further development **evidence** is needed. Both prompts have four passes on six questions.

Repair and save your work

Review Structured **Output** and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?.

Return to this week and explain what you changed.

Review Examples and **Rubric**s (ai-3/6). Goal: Distinguish a format example from approved factual evidence. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card.. Return to this week and explain what you changed.

Review Tokens and Context (ai-3/3). Goal: Account for fixed text, **evidence** and reserved **output** under a toy budget. Short practice: Can A alone justify answering 20 seats after the exception is known?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

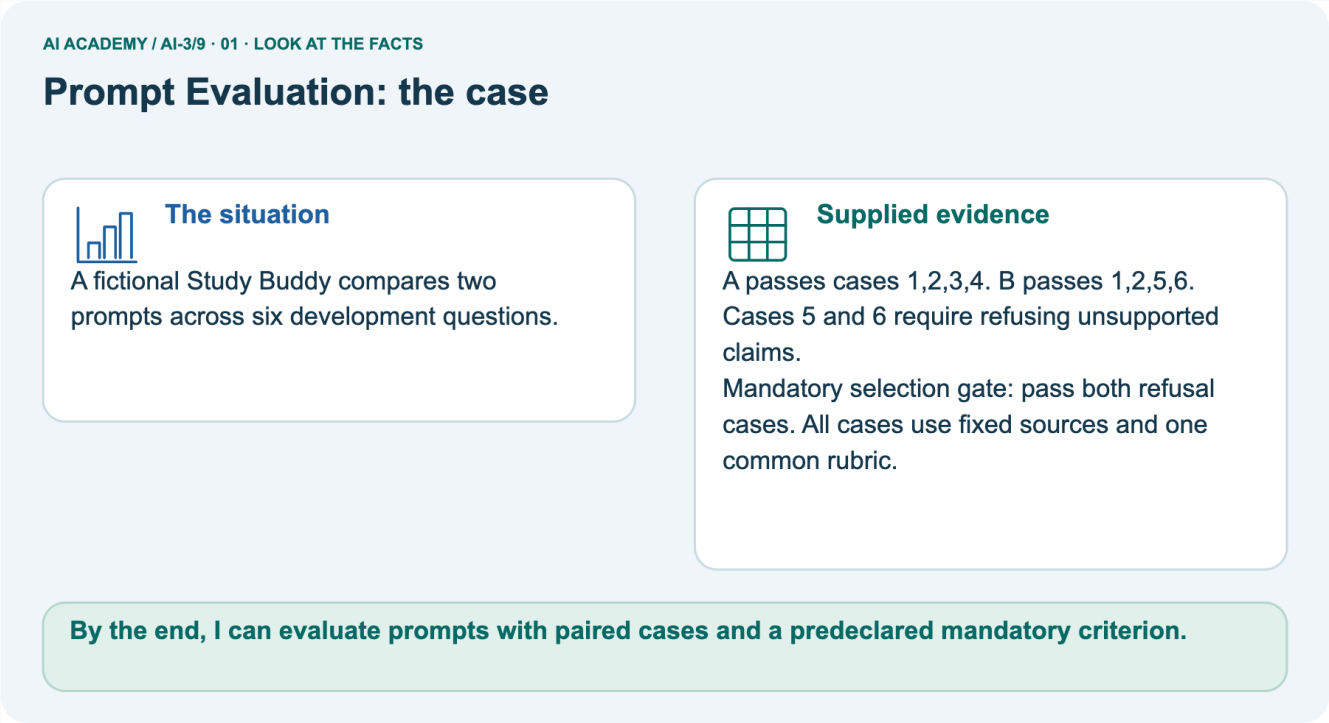
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Prompt Evaluation: the case



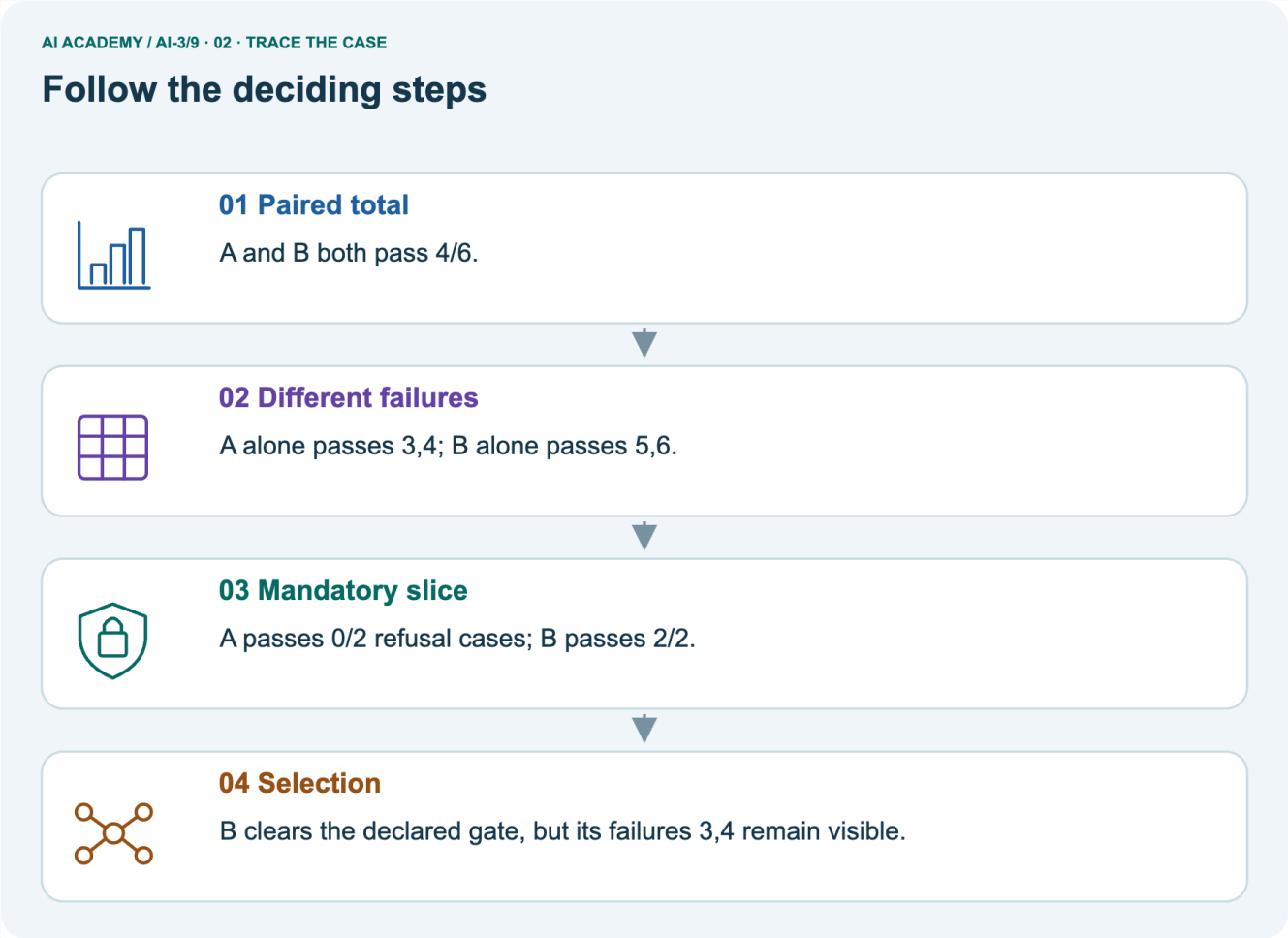
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy compares two prompts across six development questions. Supplied facts: A passes cases 1,2,3,4. B passes 1,2,5,6. Cases 5 and 6 require refusing unsupported claims.

Mandatory selection gate: pass both refusal cases. All cases use fixed sources and one common **rubric**.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

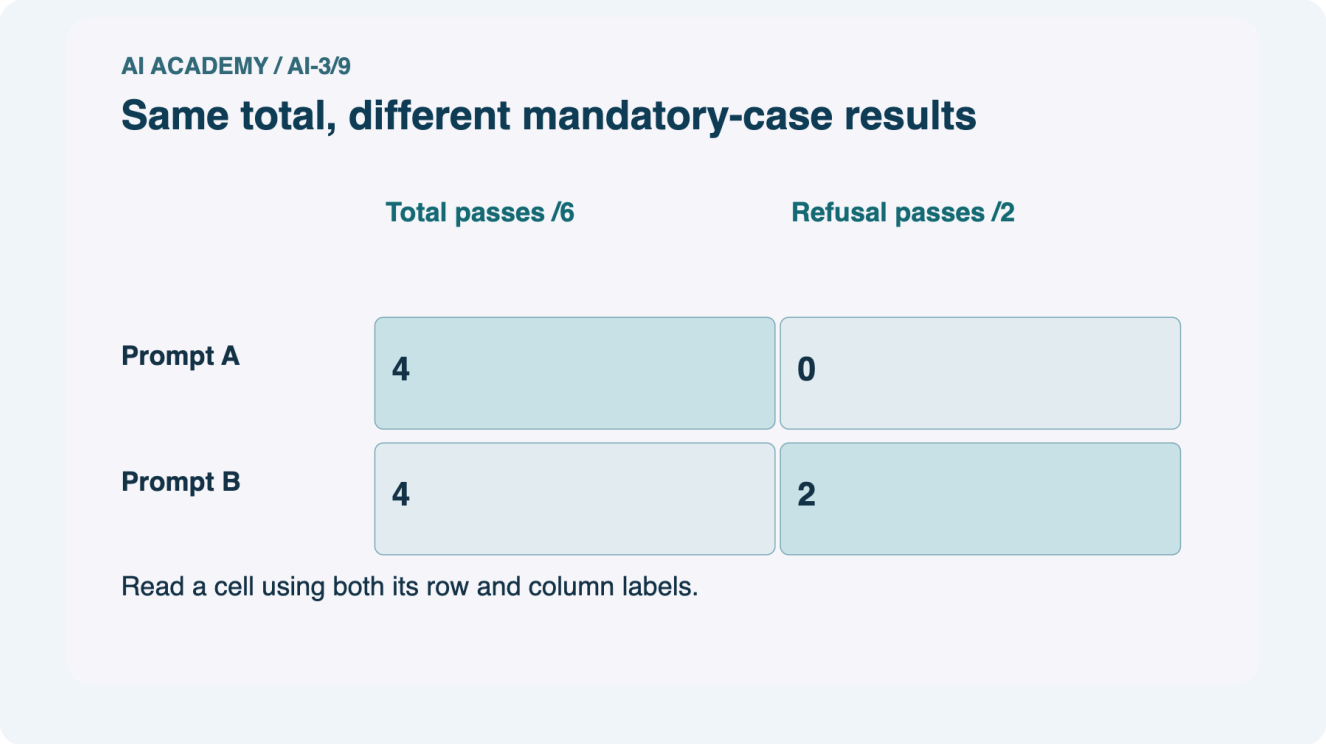


Follow the labelled picture with the matching explanation. Open the vector diagram

Paired total: A and B both pass 4/6. → Different failures: A alone passes 3,4; B alone passes 5,6. → Mandatory slice: A passes 0/2 refusal cases; B passes 2/2. → Selection: B clears the declared gate, but its failures 3,4 remain visible.

Point to each numbered stage and explain which recorded **input** it uses.

03. Same total, different mandatory-case results

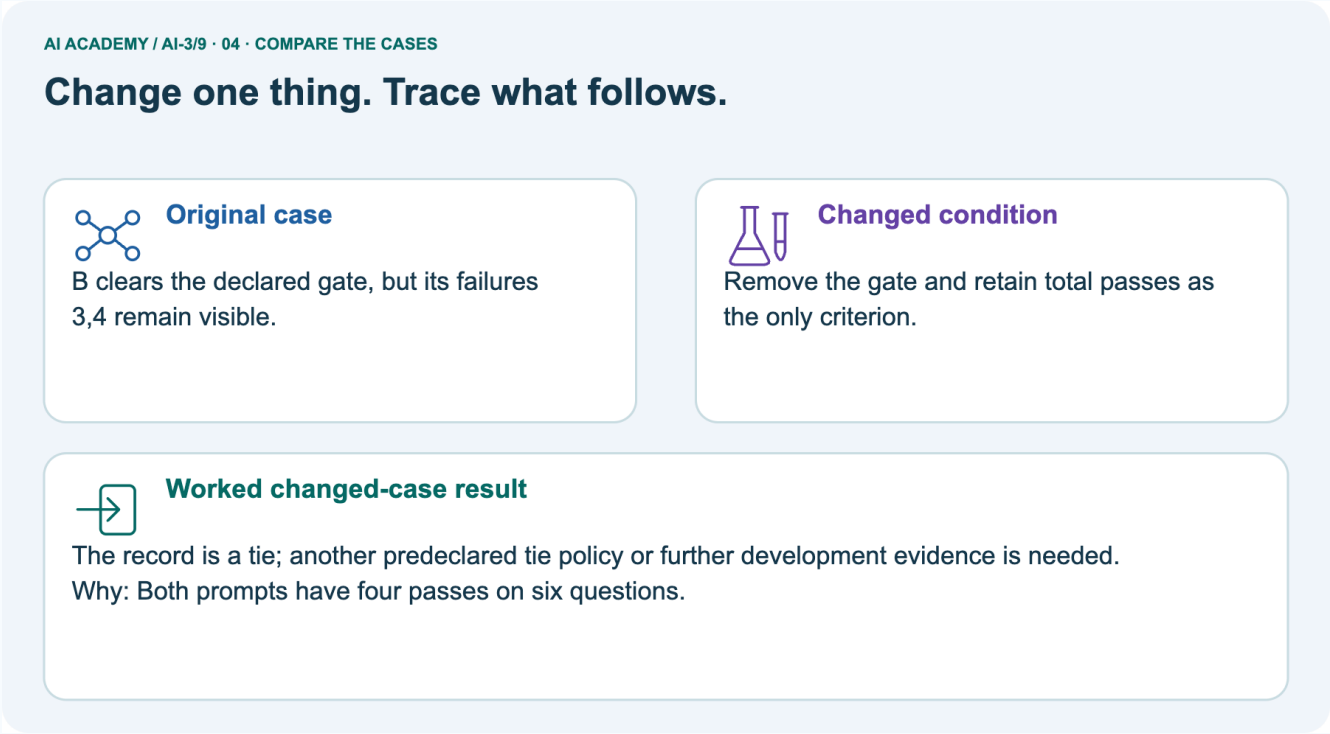


Follow the labelled picture with the matching explanation. Open the vector diagram

Same total, different mandatory-case results. Prompt A: Total passes /6 = 4, Refusal passes /2 = 0. Prompt B: Total passes /6 = 4, Refusal passes /2 = 2.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



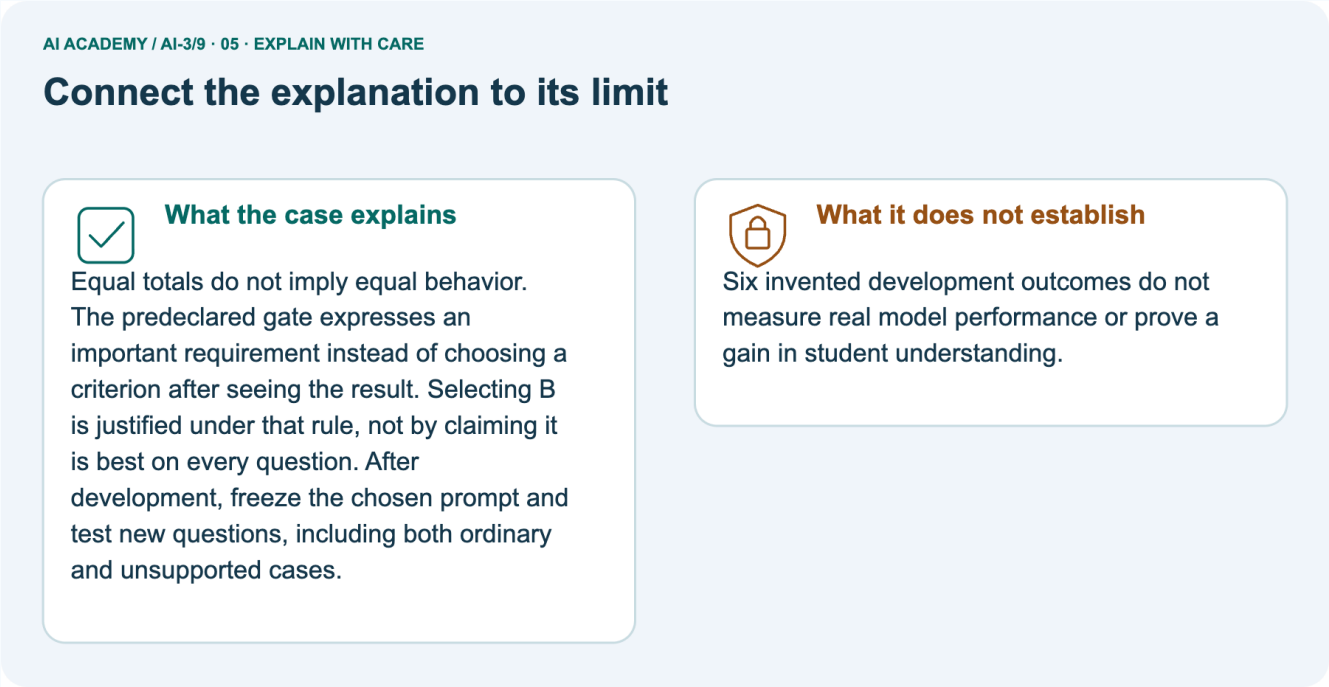
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B clears the declared gate, but its failures 3,4 remain visible. Changed condition: Remove the gate and retain total passes as the only criterion.

Expected result: The record is a tie; another predeclared tie policy or further development **evidence** is needed. Reason: Both prompts have four passes on six questions.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

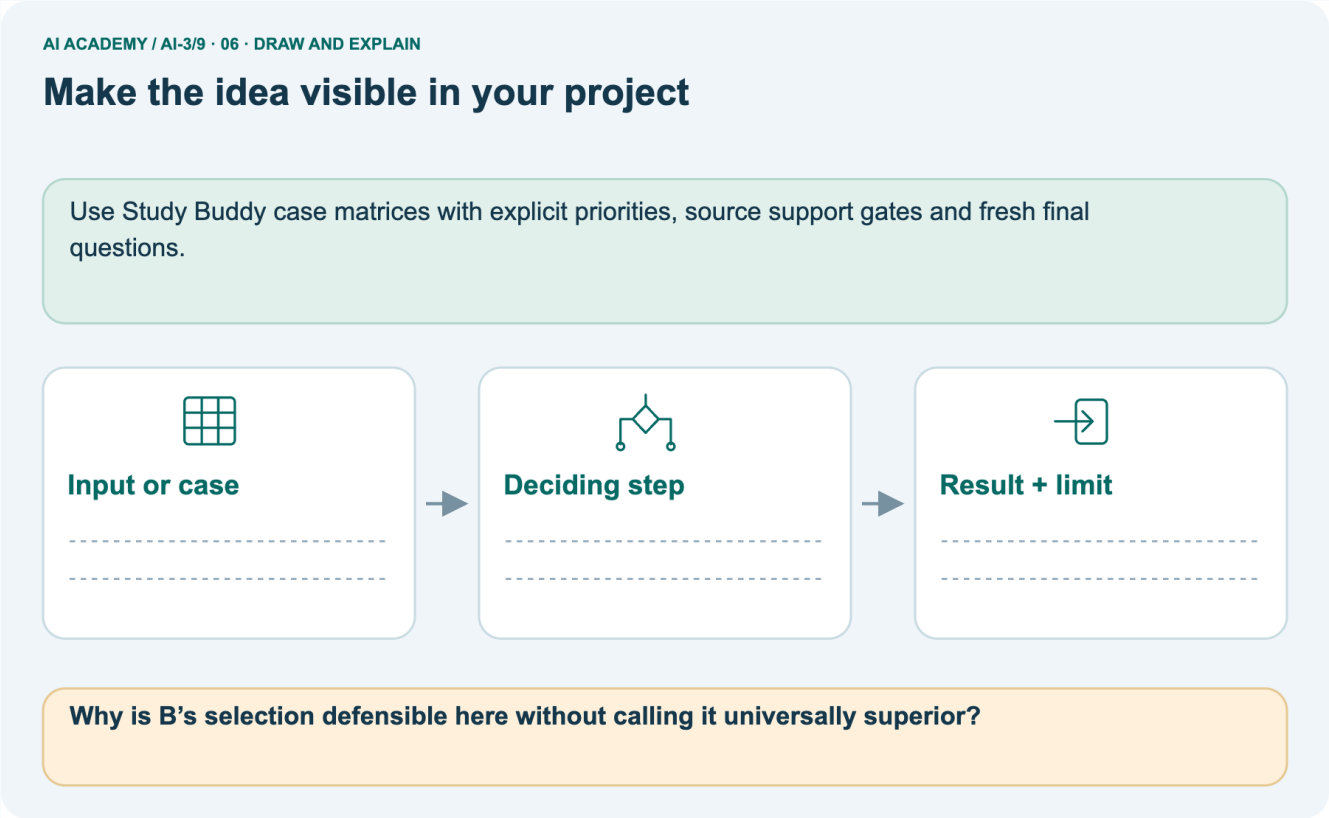
Equal totals do not imply equal behavior. The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result.

Selecting B is justified under that rule, not by claiming it is best on every question. After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

Limit: Six invented development outcomes do not measure real model performance or prove a gain in student understanding.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions. Draw your **input**, deciding step, result and limit.

Supported question: Why is B's selection defensible here without calling it universally superior?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Prompt Evaluation: the case

02. Follow the deciding steps

03. Same total, different mandatory-case results

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.